

PHYS 230 - Electricity and Magnetism

Lecture 22

Instructor: Saeed Rastgoo



**UNIVERSITY
OF ALBERTA**

Outline

5 Current, Resistance, and Electromotive Force

- Resistivity, Resistance, and Ohm's Law
- Electromotive Force

All figures in these slides are from one of the following sources, unless cited otherwise:

University Physics With Modern Physics, Young & Freedman

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Resistivity and Ohm's Law

In most metals at a given temperature **J** is proportional to the applied **E**

$$\mathbf{J} \propto \mathbf{E}$$

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- The proportionality constant ρ is called **resistivity** and is defined as

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- The proportionality constant ρ is called **resistivity** and is defined as

$$\rho = \frac{|\mathbf{E}|}{|\mathbf{J}|} \Rightarrow \rho = \frac{E}{J}$$

- The greater the resistivity ρ , the stronger $\mathbf{E}$ needed to create a certain current density $\mathbf{J}$

Resistivity and Ohm's Law

Units of resistivity

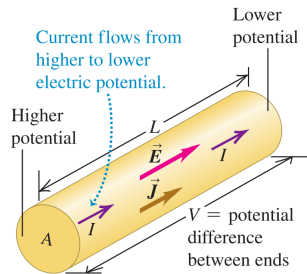
$$[\rho] = \frac{[E]}{[J]} = \frac{\frac{\text{V}}{\text{m}}}{\frac{\text{A}}{\text{m}^2}} = \frac{\text{V}}{\text{A}} \cdot \text{m} = \Omega \cdot \text{m}$$

where $\Omega = \text{V}/\text{A}$ is called an Ohm

Resistance and the Ohm's Law

Consider a wire conductor of length L , in which

Figure 25.7 A conductor with uniform cross section. The current density is uniform over any cross section, and the electric field is constant along the length.

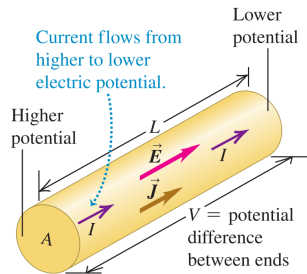


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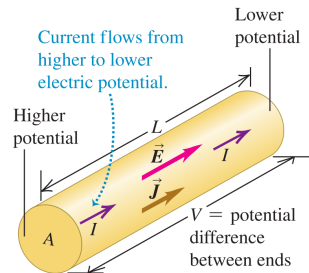


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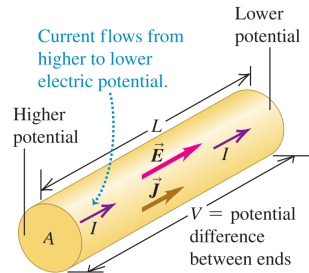


Resistance and the Ohm's Law

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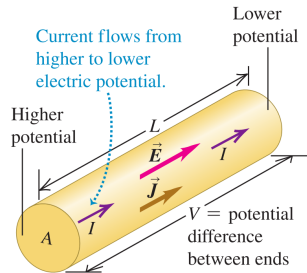
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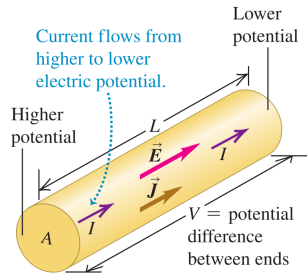
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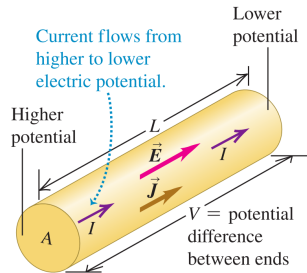
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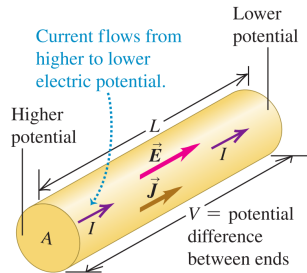
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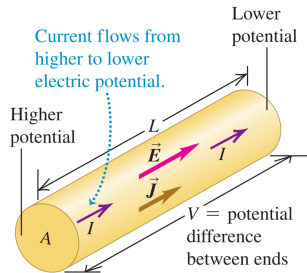
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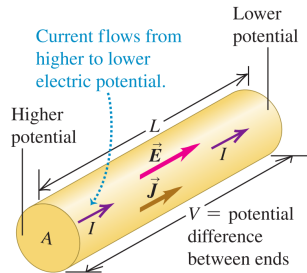
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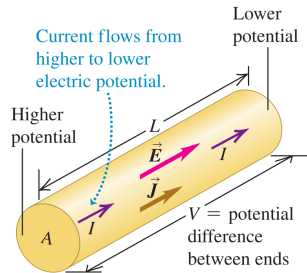
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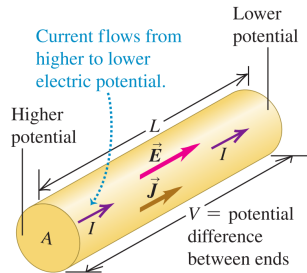
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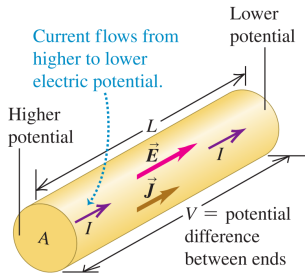
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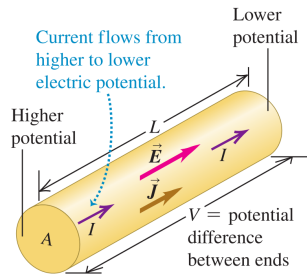


Resistance and the Ohm's Law

Replacing these into $E = \rho j$ we get

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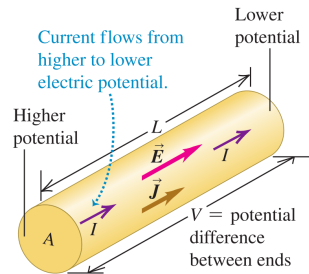


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$$E = \rho j \Rightarrow \frac{V}{L} = \rho \frac{I}{A}$$

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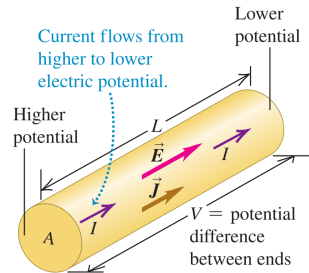


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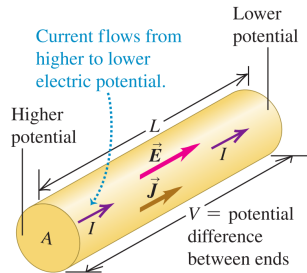
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We define the constant $\frac{\rho L}{A}$ as **resistance R**

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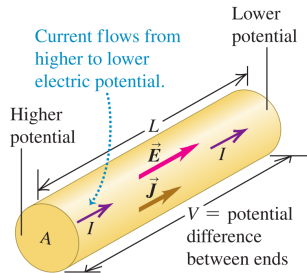
We define the constant $\frac{\rho L}{A}$ as **resistance R**

$$R = \frac{\rho L}{A}$$

And the equation in the first line becomes another form of the **Ohm's law**

$$V = RI \Rightarrow I = \frac{V}{R}$$

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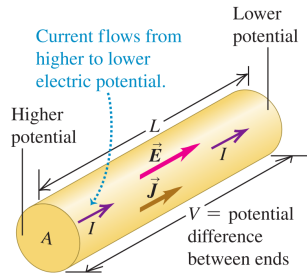
Resistance and the Ohm's Law

Equation

$$V = RI$$

defines R and is true for any conductor even when R is not constant, but when R is constant, it is called the Ohm's law

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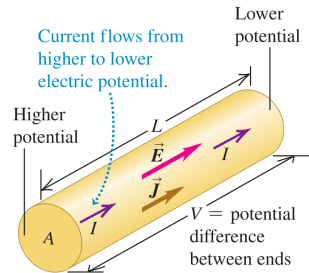
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The dimensions of resistance R is **ohm, Ω** ,

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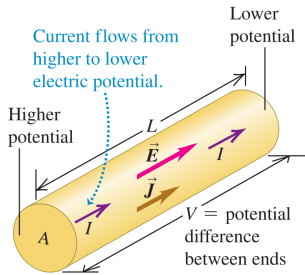
The dimensions of resistance R is **ohm, Ω** ,

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The higher the resistance, the larger V (and thus E) is needed to produce a certain current

$$R = \frac{V}{I}$$

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Exercise I

<https://epoll.srv.ualberta.ca> – Code: MBD

The copper wire in the picture has a cross-sectional area of $7.54 \times 10^{-7} \text{ m}^2$. It carries a current of 0.26

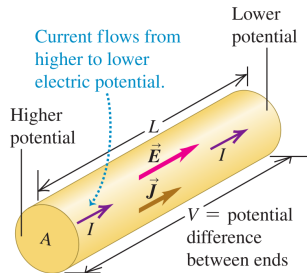
A. For copper $\rho = 1.72 \times 10^{-8} \Omega \cdot \text{m}$. Find

(a) the electric field magnitude in the wire

(b) the potential difference between two points in the wire 50.0 m apart

(c) the resistance of a 50.0 m length of this wire

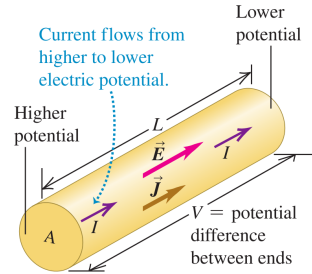
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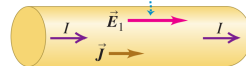
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$\vec{E}$, $\vec{J}$ and I in an Isolated Conductor

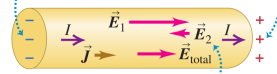
In an isolated conductor with resistivity ρ :

Figure 25.11 If an electric field is produced inside a conductor that is *not* part of a complete circuit, current flows for only a very short time.

(a) An electric field $\vec{E}_1$ produced inside an isolated conductor causes a current.

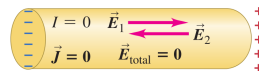


(b) The current causes charge to build up at the ends.



The charge buildup produces an opposing field $\vec{E}_2$, thus reducing the current.

(c) After a very short time $\vec{E}_2$ has the same magnitude as $\vec{E}_1$; then the total field is $\vec{E}_{\text{total}} = 0$ and the current stops completely.



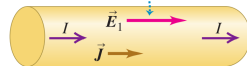
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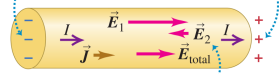
- If we establish an electric field $\mathbf{E}_1$ inside the conductor, a current begins to flow with current density $\mathbf{J} = \frac{\mathbf{E}_1}{\rho}$

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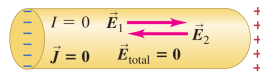


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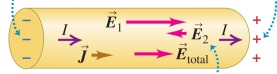
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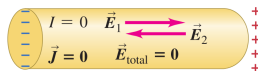


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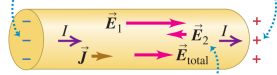
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- So a net positive charge quickly accumulates at one end of the conductor and a net negative charge accumulates at the other end
- These charges themselves produce an electric field $\mathbf{E}_2$ in the opposite direction $\mathbf{E}_1$

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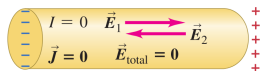


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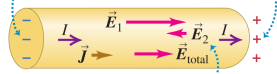
- If we establish an electric field $\mathbf{E}_1$ inside the conductor, a current begins to flow with current density $\mathbf{J} = \frac{\mathbf{E}_1}{\rho}$
- So a net positive charge quickly accumulates at one end of the conductor and a net negative charge accumulates at the other end
- These charges themselves produce an electric field $\mathbf{E}_2$ in the opposite direction $\mathbf{E}_1$
- This causes the total electric field $\mathbf{E} = \mathbf{E}_1 + \mathbf{E}_2$ and hence the current to decrease

Figure 25.11 If an electric field is produced inside a conductor that is *not* part of a complete circuit, current flows for only a very short time.

(a) An electric field $\vec{E}_1$ produced inside an isolated conductor causes a current.

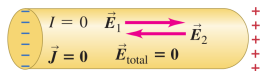


(b) The current causes charge to build up at the ends.



The charge buildup produces an opposing field $\vec{E}_2$, thus reducing the current.

(c) After a very short time $\vec{E}_2$ has the same magnitude as $\vec{E}_1$; then the total field is $\vec{E}_{\text{total}} = 0$ and the current stops completely.



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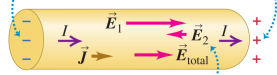
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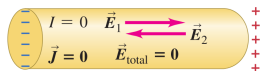


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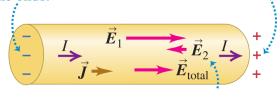
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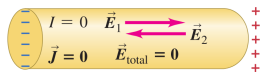


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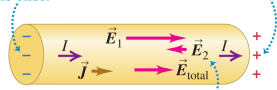
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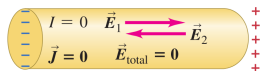


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- When U or equivalently V increases, q goes from a lower potential to a higher potential, against the usual case in conductors

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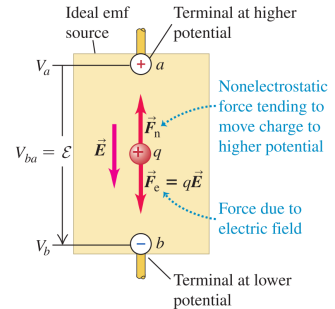
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Every complete circuit with a steady current must include a source of emf

Source of emf

Figure: a battery; maintains a potential difference $\mathcal{E} = V_+ - V_-$ between conductors a and b

Figure 25.13 Schematic diagram of a source of emf in an “open-circuit” situation. The electric-field force $\vec{F}_e = q\vec{E}$ and the nonelectrostatic force $\vec{F}_n$ are shown for a positive charge q .



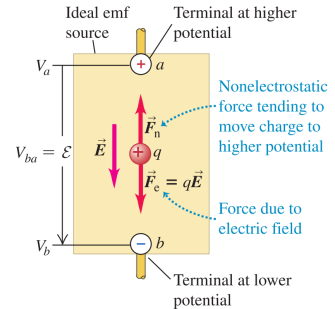
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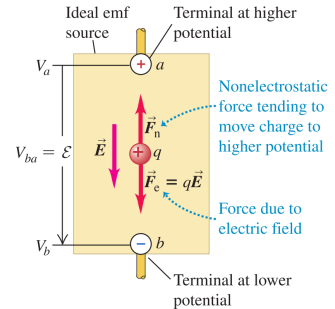
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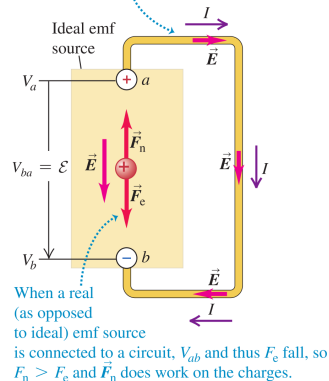
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Associated with $\mathcal{E}$ is an electric field $\mathbf{E}$, both inside and outside the battery (in the parts of circuit outside the battery)

Figure 25.14 Schematic diagram of an ideal source of emf in a complete circuit. The electric-field force $\vec{F}_e = q\vec{E}$ and the nonelectrostatic force $\vec{F}_n$ are shown for a positive charge q . The current is in the direction from a to b in the external circuit and from b to a within the source.

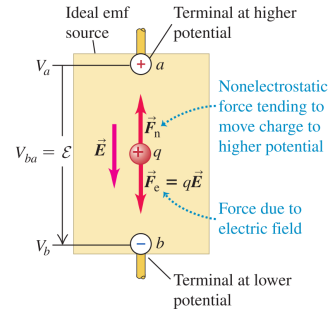
Potential across terminals creates electric field in circuit, causing charges to move.



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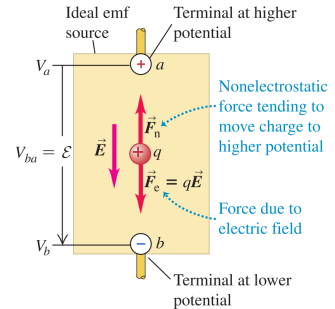
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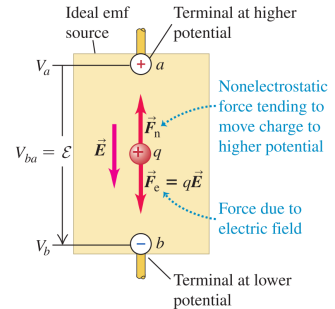
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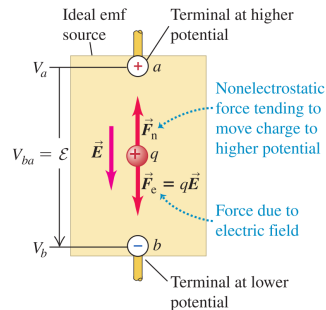
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When $\mathbf{F}_n$ moves q from $-$ to $+$, the electric potential energy of q increases since

$$U_+ - U_- = q(V_+ - V_-) \Rightarrow U_+ = qV_+ + U_-$$

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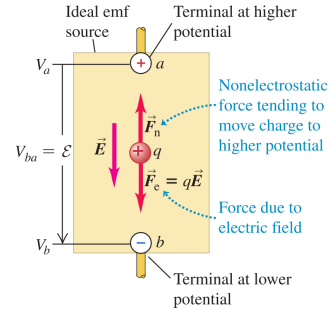


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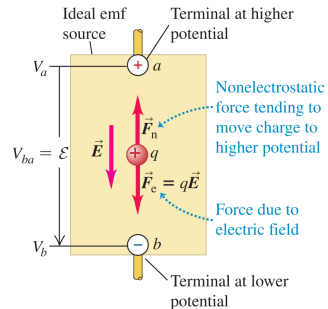


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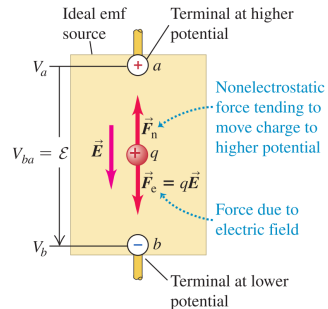
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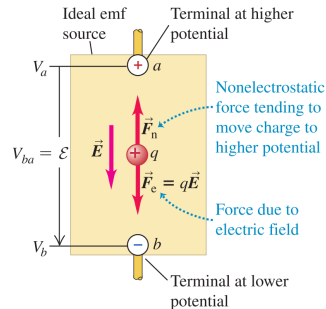
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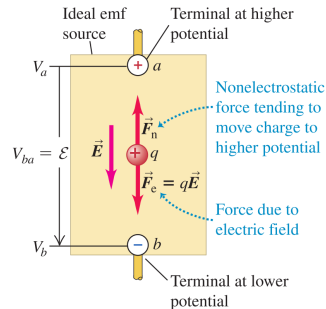
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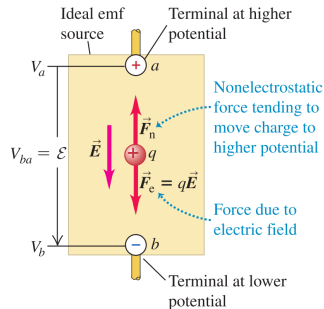
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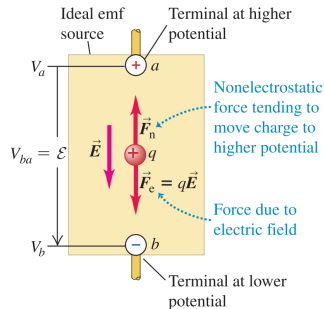
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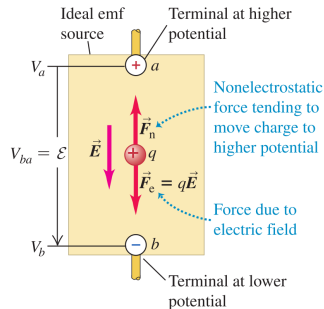
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LHS is energy (or work) per unit charge in moving q : definition of $\mathcal{E}$

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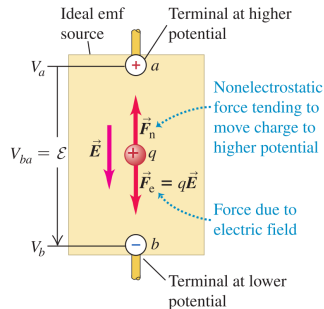
Dividing by charge q we get

$$\frac{W_{n(-+)}}{q} = - \int_{-}^{+} \mathbf{E} \cdot d\mathbf{r} \Rightarrow \frac{W_{n(-+)}}{q} = V_{+} - V_{-}$$

LHS is energy (or work) per unit charge in moving q : definition of $\mathcal{E}$, so

$$\mathcal{E} = V_{-+} \quad (\text{or } \mathcal{E} = V_{ba})$$

Figure 25.13 Schematic diagram of a source of emf in an “open-circuit” situation. The electric-field force $\vec{F}_e = q\vec{E}$ and the nonelectrostatic force $\vec{F}_n$ are shown for a positive charge q .



When the emf source is not part of a closed circuit, $F_n = F_e$ and there is no net motion of charge between the terminals.